Keshav Choudhary

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.08
Arihant Public School, Ladpura, Kota, Rajasthan	Class XII (CBSE)	2022	86.40%
Kendriya Vidyalaya No.1, Kota, Rajasthan	Class X (CBSE)	2020	86.80%

WORK EXPERIENCE

Teaching Assistant for Machine learning course (ME6040) under Dr. Venkatesham B.

Semester-long Internship | Carrier India LTD.

- Performed modal analysis of Air management System of outdoor unit using Finite Element Analysis (FEA) in Nastran, extracting natural frequencies and mode shape of the structure.
- Prepared analysis ready assembly (Geometry simplification, meshing and creating interaction).
- Identified natural frequencies near motor frequencies, indicating potential noise and fatigue risks.
- Passed the modes shapes and frequencies to be further analysis.

PROJECTS

Optimization-Based Characterization of Elastoplastic Materials using Python & MATLAB (under Dr. Sai Siddardh)

- Developed a programme to predict material properties of a **Ramberg-Osgood** material given its stress-strain graph.
- Used **Differential evaluation algorithm** to reach the **global minima** then uses curve fit method to get near exact values.

Shortest Pathfinding System

Using Dijkstra's Algorithm this project computes the shortest path to a given target city from a given source city.

Developed a solver for Beam Bending Analysis using MATLAB (under Dr. Chandra Prakash)

- Accepts **user defined loading and support** conditions and solves for bending and stress distributions using **FEA.**
- **Validated** the code for various user defined input conditions by comparing with analytical solutions.

Machine Learning-based Vibration Characterisation of a 1050 W BLDC motor (Under Dr. Venkatesham B)

- Successfully validated the methodology for calculating Natural frequencies and mode shapes.

Developed Perfectly Matched Layers for absorbing Incident Waves

- Created surrounding subdomains with **proportional damping** to reduce reflections using **Python scripting**.
- Successfully reduced the reflections by **99 %** enabling near reflection-free-boundaries for wave propagation**.**

Neural Network-Based Stress Intensity Factor Prediction

- Generated SIF data by varying geometric parameters and trained an **Artificial Neural Network model** for prediction.

CONFERENCES

Presented a **Conference Paper** on **Physics Informed Neural Networks** at 35th Annual Conference ansd exhibition on Non-Destructive Evaluation and Enabling Technologies 2025 organised by Indian Society for Non-Destructive testing.

RELEVANT COURSES

Machine Learning Courses

Supervised Machine Learning and Advanced Learning Algorithms.

Mathematics Courses

Numerical Methods, Differential Equations, Probability, Linear Algebra, calculus I and calculus II.

Mechanical Theory Courses

Finite Element Method, Solid mechanics, Advanced Finite Element Method, Design of Machine Elements, Modelling and Simulations, Engineering Mechanics, Kinematics and Dynamics of Mechanisms, Manufacturing Science, Computational Fluid Dynamics, Power and Refrigeration Systems, Fluid Mechanics, Thermodynamics and Heat and Mass transfer.

SKILLS

Programming and Analysis: C++, Python, MATLAB and C.

Machine Learning

Linear Regression, Logistic Regression, Decision Trees, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), K-Means Clustering

Finite Element Analysis (FEA): NASTRAN, Abaqus CAE and Ansys Workbench.

POSITIONS OF RESPONSIBILITY

Placement Coordinator at Placement Cell, IIT Hyderabad

Core Member of Torque, IIT Hyderabad

EXTRACURRICULAR

Represented IIT Hyedrabad in inter IIT Sports meet 2023 at IIT Gandhinagar.